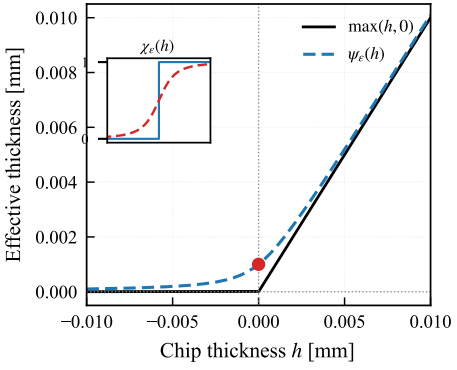
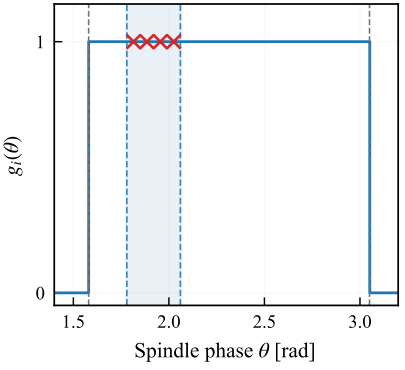
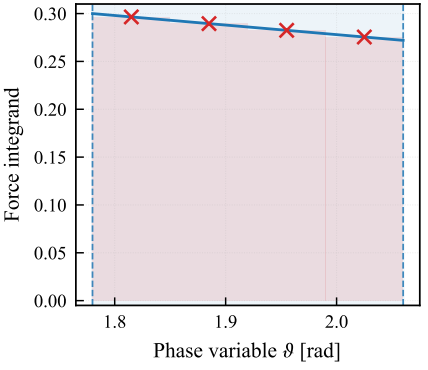

(a) Chip-loss
Effective thickness [mm]
0.010
0.008
0.006
0.004
0.002
0.000
$\chi_\epsilon(h)$
1
0
$\max(h, 0)$
$\psi_\epsilon(h)$
−0.010
−0.005
0.000
0.005
0.010
Chip thickness h [mm]


(b) Engagement
$g_i(\theta)$
1
0
1.5
2.0
2.5
3.0
Spindle phase θ [rad]


(c) Active-interval quadrature
Force integrand
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1.8
1.9
2.0
Phase variable ϑ [rad]